Queen Alexandra's Birdwing Butterfly (*Ornithoptera alexandrae* Rothschild).

A review



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Introduction

The Queen Alexandra's Birdwing butterfly (*Ornithoptera alexandrae*, Rothschild) is, justifiably, an insect that deserves a lot of attention both because of its size and its localised habitat, which contributes to its perceived rarity. QABB, as it will be referred to throughout this review, is recognised as the world's largest butterfly with the female wingspan ranging up to 30cm and male wingspan up to 24cm. Because of this, it came under a lot of pressure from collectors in the early to mid parts of the last century and has been constantly under pressure from shifting and permanent cultivation, logging, volcanic activity and changes in weather patterns. Much has been said and written about QABB and, because of it's high profile, many people from different walks of life have become involved in one way or another (e.g. farmers, land owners, collectors, traders, conservationists, scientists, donors and politicians). There has therefore been an accumulation of facts, conjecture, informed and uninformed statements that have been passed down over time. The purpose of this review is to examine what is known about QABB and to bring together the various ideas on its conservation and to suggest a way forward which can be adopted by the various stakeholders to ensure continuity in conservation for the future. The review is based on various papers, reports, file notes and some interviews with officers who are working or have worked on the study and conservation of QABB. Unfortunately the author has not sighted some important reports, and has had to rely on other reports, reviews and papers to glean the information contained in those reports.

Alfred Meek first collected QABB near or around Kokoda in January 1906 and it was named by Lord Rothschild (Meek's employer) in 1907 in honour of Queen Victoria's sister, Alexandra. Meek had to shoot the female specimen down with a shotgun as it was flying high in the canopy. He did not specify the area very clearly other than he was at "Biagi, 5000ft high, at the head of the Mambare River" The Biagi tribe were said to be on the upper slopes of the mountains "plainly visible from Kokoda". With a locality description like this it is difficult to know exactly where the specimen was collected other than it was not at Kokoda itself. Mercer (1999) says that the butterfly has never been recorded in the Kokoda area since, although Parsons (1992) reports unconfirmed sightings in the area during the 1980's. Meek also caught the first male specimen about a year later (1907) and his locality for this specimen was even vaguer than for the first female: a "position under the Owen Stanley ranges", which has been presumed to be around Popondetta. The two type specimens for this species therefore have very poorly recorded locations. By the end of 1908 Meek had collected perfect specimens of both male and female QABB and had also found the larvae and pupae.

Between the early collections and the 1960s, a general outline of the life cycle of QABB was established. Along with most of the other birdwing butterfly species, QABB became highly collectable and various collectors and dealers trained local people to collect and rear the larvae and pupae so that perfect specimens of the butterflies could be obtained. This caused concern amongst some entomologists who recognised that, because the localised nature of QABB's distribution, there was a danger of the species being over-collected and the butterfly becoming extinct. Legislation was therefore brought in to protect seven species of birdwing butterflies under the Fauna Protection Act (1968), which was superseded by the 1974 Fauna (Protection and Control) Act. This act prohibits the killing, selling, buying or possession of a protected species. The International Union of Conservation of Nature (IUCN) lists QABB as endangered. QABB is

also said to be on Appendix 1 of the Convention on International Trade of Endangered Species (CITES). Various authors (e.g. Mercer, Orsak, Douglas) state that QABB was put onto Appendix 1 during the 1980's, several authors sighting Collins and Morris 1985. However in 1992, Parsons, who originally suggested and campaigned that QABB should be on Appendix 1, says: "This recommendation was later adopted in print by Collins and Morris (1985) but, to date, *O. alexandrae* remains on Appendix II". According to the CITES website QABB is on Appendix 1. Despite the legislation, many collectors and dealers have been caught with QABB and, given the high price paid for an individual specimen in Europe and America (over $1000); the temptations are high for illegal trading.

There have been a number of studies of the biology, ecology, distribution, habitat and conservation of QABB and various strategies have been developed. The history of these studies can be summed up as follows:

a. 1970-71. Ray Straatman was commissioned by the PNG Government to carry out the first detailed study of QABB. He produced sketch maps of three areas near Popondetta where the butterflies could be found, discovered a population between Afore and Sila, made initial descriptions of the habitats and produced a study on the life cycle of QABB.
b. 1979-1996. Michael Parsons was at the Insect Farming and Trading in Bulolo between 1979 and 1983. During this time he made several visits to Oro Province to survey the distribution of QABB and other butterflies and wrote a series of reports and papers between 1980 and 1984 (7 relevant to QABB). His four main reports to the PNG Division of Wildlife covered: (i) continuation of Straatman's work, giving new localities of QABB populations on Popendetta plain (ii) oil palm development, concluding that it is incompatible with QABB presence in same area (iii) the likely effects of logging on Popendetta plain by Kumusi Timbers (iv) proposals for three conservation sites known to have large QABB populations: Siai / Mangure, Girigirita / Ongoho, and Huhuru / New Warisota. Parsons also made periodic visits to PNG during the rest of the 1980's and produced a number of high profile reports and papers between 1985 and 1990. In 1991 he carried out a two-month survey of QABB on the Popondetta plain with a short visit to Afore/Kawowoki on the nearby Managalase Plateau. He then produced a comprehensive review of the state of knowledge of QABB in 1992 and a paper proposing revisions of the genus of *Aristolochia* (food vines of several of the birdwings) in New Guinea and Australia. In 1998 he produced "The Butterflies of Papua New Guinea: Their Systematics and Biology" a 928 page book covering 958 butterfly species with photos of 3000 specimens and 400 colour photos detailing aspects of life history and ecology. Michael Parsons is regarded as the world authority on QABB.
c. 1984. W. Moi produced reports to the Office of Environment and Conservation on habitat conservation.
d. 1992. Larry Orsak produced, at the request of the PNG Department of Environment and Conservation (DEC), a five-year management plan for QABB. This covered (i) a summary of the existing knowledge of QABB, speculating the reasons why it is a rare insect (ii) organisations participating in the conservation of QABB (iii) a framework for a conservation programme (iv) strategies for success of a conservation programme (v) biological considerations in the conservation of QABB (vi) a management action plan with key outcomes
e. 1998. Simon Saulei carried out some limited work on QABB habitat.

f. 1991-1994. DEC, Chris Mercer, and students from Unitech carried out a number of surveys to delineate further the distribution of QABB. The surveys resulted in the known range of QABB being increased by 50%, and suggested that Managalase Plateau has the main QABB population.
g. 1995 – 1999. The Oro Conservation Project was instigated under a component of a World Bank loan for the expansion of smallholder oil palm blocks in Oro Province. After discussions, the QABB conservation component was transferred to a grant program under AIDAB (now AusAid) auspices. The project pooled the resources of the AusAid Project, DEC and the Oro Provincial Government in promoting research and conservation of QABB. The project covered various areas of research on QABB: (i) distribution (ii) monitoring (iii) habitat creation and enrichment planting (iv) captive rearing (v) biology and ecology (vi) food plant ecology and taxonomy (vii) the effects of environmental disturbance and (viii) conservation. Chris Mercer, the Principal Scientist on the project, wrote a final report in 1999.
h. Since the end of the above project there has been little work done on QABB. A workshop of interested parties was held in 2001 to discuss future conservation needs but little transpired from that. Also in 2001, John Douglas produced an assessment for DEC of further actions needed to protect QABB. OPRA and HOP have been involved in carrying out monitoring surveys, working on the propagation and planting out of the food vine for QABB and have been helping the Government Offices (the Provincial Wildlife Officer and the DEC representative for Oro Province) in encouraging and organising the setting up of Wildlife Management Areas.

Life cycle

The whole life cycle

Total life cycle takes 9 ½ weeks, just less than five months
QABB activity is seasonal
Caterpillars are most frequently found during the wet season when the vines put on new leaves
Opposite for adult butterflies – seldom seen during the wet season, common from May to October
Mortality factors include a hymenopteran parasitoid, possible nematode infections, spiders and possibly birds

The imagos (adult butterflies)

Males are metallic- blue/green with a wing span of 22-24cm
Females are brown and orange/yellow with a wing span of 27-29cm
QABB from Managalase Plateau are much larger than those from Popendetta Plain
Adult QABB are strong fliers and cover long distances
Males and females have a courtship dance prior to mating
Both sexes visit a number of flowering trees, vines and bushes to obtain nectar
Trees include Kwila, Teak and *Kleinhovia* sp
Vines include Morning Glory and Tobatoba
Bushes include Hibiscus, purple Ipomea, Bougainvillea and Ixora, as well as coffee (during flowering)

The ovae

Egg stage – 12-14 days,
Egg deposited either on underside of leaf of food plant vine, or under leaf of the supporting tree

The larvae

6 larval instars – 72-75 days,
3-day pre-pupal stage
Males generally go through 5 instars and female 6
Those raised on highly nutritious vines often skip one of the later instars
First and second instar caterpillars can only chew tender new leaves
1st and 2nd instar caterpillars feed on new leaves and the growing tip of vines
As caterpillars get older they progressively move down the vine to feed on older and tougher leaves

The pupae

pupal stage of 49 days
4 pupation sites

1. under a vine just below the tree canopy
2. under a leaf in crown of tree
3. under a vine leaf close to the last feeding site, usually within five metres of ground
4. pupate under a leaf in nearby bushes, up to 10 m away from vine

Distribution

Distribution on Managalase Plateau and surrounding coastal hills

Butterfly monitors set up at Kawowoki , Tabuane, Tahama and Toma
Have all found immature stages of QABB
Caterpillars also sighted at Koruwo, Siribu, Semehara and at intervening sites across plateau
Following villages found in areas of QABB habitat and recommended for conservation:- (1) Semehara, (2) Toma, (3) Gewoia, (4) Korala, (5) Umbara, (6) Dareki, (7) Sigara, (8) Kawowoki, (9) Afore, (10) Yoivi, (11) Yoivi, (12) Kweno, (13) Tabuane, (14) Dea, (15) Koruwo, (16) Tahama, (17) Siribu

Distribution at Collingwood Bay

Lowland form of *P. dielsiana* foodplant widely distributed in Wanigela area of Collingwood Bay

QABB not found in this area
Vines grow in abundance on nutrient rich soils derived from volcanic ash, in similar conditions as in Popendetta Plain and Managalase Plateau

Cape Ward Hunt

2 QABB found at Taututu along Mambare River
Very few vines located and no immature stages of QABB sighted
Seems to be a very small, isolated population

Popendetta Plain

Mostly a mosaic of forest fragments interspersed with *Imperata* grassland
QABB habitat remains of the slopes of Mount Lamington and to the north and west of the Plain
Oro Conservation Project has focused its research and conservation efforts at three sites:-
(1) the proposed Voivoro Wildlife Management Area at Ondahari (approx 400 ha) in north-west
(2) the Hombareta Wildlife Management Area (approx 130 ha) in the centre
(3) the Gewota / Sananada proposed WMA (approx 800 ha) in north-east, near to coast

Enrichment planting with vines has been going on at Voivoro and Hombareta since early 1970
Voivoro is one of best sites for QABB
Hombareta does not appear to have QABB any more – is now completely surrounded by oil palm
130 ha Hombareta WMA appears to be too small to sustain a continuous QABB population
Appears to be a lower critical size to a reserve, below which the habitat can no longer sustain QABB habitats

Effects of environmental disturbance on QABB habitat

shifting cultivation results in forest clearance
has intensified over past 20 years with increasing population
many areas in Managalase Plateau are running out of land, resulting in shortened rotation times
vegetation becomes degraded and QABB vines do not grow

Habitat

Pararistolochia dielsiana (??) vines have a far greater distribution than QABB, but QABB is restricted to 3 areas in Oro Province
Thought to be the quality of the volcanic soils promote vigorously growing vines at high density

Chemical cues determine the initial stages of oviposition behaviour
Perhaps combined with a visual cue
Vine density said to be important factor in inducing oviposition:-

100+ vines / ha – excellent habitat
50-100 vines / ha – good habitat
50-30 vines / ha – frequently utilised
10-30 vines / ha – occasionally utilised
less than 10 vines/ ha – seldom utilised

Forest structure also important in determining if vines are to be utilised, even with good vine density
Primary and secondary forest sites utilised by QABB show openness
5 metre + gap between canopy trees and next tree storey

Interaction with the host plants

Larval food plant Pararistolochia dielsiana

Adult nectar food plants

Behaviour

Female QABB will not oviposit on a vine that already has an immature stage of either its own species or *Ornithoptera priamus* already on it (Straatman and Parsons say the opposite: can get many larvae on one vine but none on vines nearby)
Female QABB will only deposit one egg per vine
Growth phase of vine is important in inducing oviposition
Female will deposit egg within 2 metres of actively growing shoot
Vines with no growing shoots are usually not chosen as oviposition sites

Attempts at conservation

Legislation

Conservation areas

Oil palm results in the complete destruction of QABB habitat in areas where it is grown.
Impossible for vines to grow in oil palm agroecosystem (??)
Proposed that reserve areas of forest should be left throughout HOPPL estates
One such reserve was 100 ha forest at Lejo Forestry Station
This area was mostly planted with oil palm by 1995
Before VOP blocks are established in forest areas inspections should be made to determine if areas are significant QABB habitats
Areas with good vine density should be preserved

Commercial logging commenced on the Popendetta Plain in early 1980s
No commercial logging yet on the Managalase Plateau or surrounding coastal hills

1986 Oro province Forestry Plan considers most of the province for logging
Logging opens up previously inaccessible areas, resulting in QABB habitat fragmentation and reduction
Strict adherence to the Logging Code of Practice including a 40 year regeneration period and the provision of an adequate system of reserves is necessary if QABB populations are to be sustained

Captive breeding

Captive breeding studies

Flight cage and shade house fabricated in Lae and erected at Afore
Both semi-circular in profile
Flight cage is 20m long, 4m wide and 3.5m high, covered with 2 x 40% white shade cloth
Shade house only 7.5m long, covered with 40% green shade cloth
Shade house used to study feeding behaviour of QABB, to determine life cycle lengths, etc
Difficult to get QABB to oviposit in the cages and breeding has not been possible

Propagation of vines

Techniques for raising food plant cuttings have been developed
3 months after planting of cuttings the vines are ready to plant out in the field
Even with best nursery techniques and use of hormone rooting gel, only about 2/3 of cuttings will strike and grow through to planting
Further mortality occurs after planting out, so that only 50% of cuttings will develop into mature vines

Been postulated that enrichment planting with food plant vines should increase the attractiveness of an area for QABB, and boost its carrying capacity
Has been done by several landowners with good results
Community school butterfly gardens have also been developed at (1) Afore, (2) Aguido, (3) Koruwo, (4) Gewoia, (5) Dea, (6) Umboworo, (7) Sakarina, (8) Bagou, (9) Ahora, (10) Ondahari

Current situation

Suggestions on future work

Conclusions

Bibliography